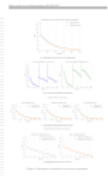




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Under review as a conference paper at ICAIS 2025

TRAIN FOR THE WORST, PLAN FOR THE BEST: ENHANCING TOKEN ORDERING IN MASKED DIFFUSIONS

Anonymous authors
Paper under double-blind review

ABSTRACT

Masked diffusion models (MDMs) have emerged as a powerful paradigm for generative modeling over discrete domains. However, their training often involves solving computationally intractable problems, while their inference capabilities remain underutilized. In this work, we propose to enhance the performance of MDMs by introducing adaptive inference strategies that allow for dynamic token ordering during decoding. We demonstrate that by sidestepping computationally heavy subproblems, pretrained MDMs can achieve significant performance improvements on complex tasks such as logic puzzles. Our experiments show that adaptive inference boosts Sudoku solving accuracy from less than 7% to approximately 90%, even outperforming autoregressive models with significantly more parameters. This work opens new avenues for leveraging the strengths of MDMs in discrete generative tasks.

1 INTRODUCTION

Masked diffusion models (MDMs) have gained traction for their ability to model complex data distributions in discrete spaces, yet they face significant challenges during training and inference. In particular, the computational complexity of their training processes and the limitations of fixed decoding strategies can hinder their usability in practical applications. Existing literature emphasizes the efficacy of autoregressive models (ARMs) for structured tasks, yet fails to explore inference strategies that exploit the flexibility of MDMs during inference. This paper addresses this gap by proposing adaptive token ordering strategies, which enhance MDMs' performance on complex generative tasks, particularly in solving logic puzzles. Our findings reveal that inference significantly improves accuracy and opens pathways for further research in generative modeling.

2 RELATED WORK

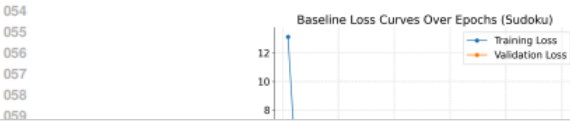
The challenges associated with MDMs include computational inefficiencies and suboptimal decoding strategies. Prior works have demonstrated the success of ARMs in structured tasks (Zheng et al., 2024; Kim et al., 2025), but lack a comprehensive analysis of adaptive inference techniques. Recent advancements in adaptive inference-time computation have shown promise in enhancing generative models (Manvi et al., 2024). Our study builds on these foundations, employing insights from adaptive sampling frameworks to optimize token selection and improve decoding efficiency in generative modeling (Lee et al., 2023; Gwak et al., 2025).

3 METHOD

We introduce adaptive inference strategies to dynamically adjust token ordering during the decoding phase of MDMs. This method leverages contextual information to prioritize token prediction, enabling the model to avoid computationally intensive subproblems. By evaluating the performance of different token sequences based on prior context, our approach enhances the model's ability to generate coherent outputs efficiently. This strategy is grounded in the theoretical framework of energy minimization (Chen et al., 2025), where optimal token arrangements can be derived to minimize generation errors.

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Under review as a conference paper at ICAIS 2025



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 Summary

This paper introduces an adaptive inference strategy for Masked Diffusion Models (MDMs), aiming to enhance their performance on discrete generative tasks by dynamically adjusting the token ordering during the decoding process. The core idea is to leverage contextual information to prioritize token predictions, thereby avoiding computationally intensive subproblems and improving overall efficiency. The authors ground their approach in the theoretical framework of energy minimization, suggesting that optimal token arrangements can be derived to minimize generation errors. Specifically, they propose an energy function that quantifies the 'cost' of a token arrangement, incorporating both local and global potentials to capture context and coherence. The method is evaluated on Sudoku puzzles, where the adaptive inference strategy significantly improves solving accuracy compared to a standard MDM. The authors demonstrate that by dynamically adjusting the token order, the model can achieve substantially higher accuracy. While the paper presents a promising approach to improving the efficiency of MDMs, it suffers from several critical limitations that hinder its overall impact. The lack of detailed methodological explanations, limited experimental validation, and poor writing quality make it difficult to fully assess the significance of the proposed method. Despite these shortcomings, the core idea of adaptive inference for MDMs represents a potentially valuable contribution to the field of generative modeling, warranting further investigation and refinement.

 Strengths

The paper's primary strength lies in its introduction of an innovative adaptive inference strategy for Masked Diffusion Models (MDMs). The core idea of dynamically adjusting token ordering during decoding, based on contextual information, is a novel approach that has the potential to significantly enhance the efficiency and performance of MDMs on discrete generative tasks. This is particularly relevant given the computational challenges associated with training and inference of MDMs, as highlighted in the introduction. The authors' grounding of their method in the theoretical framework of energy minimization provides a solid theoretical foundation. By defining an energy function that quantifies the 'cost' of a token arrangement, they provide a principled way to guide the adaptive use of both local and global potentials in the energy function demonstrates a thoughtful consideration of the contextual and coherent aspects of the task. Furthermore, the empirical results on Sudoku puzzles, while limited in scope, provide compelling evidence of the effectiveness of the proposed method. The substantial improvement in solving accuracy, from 6.8% to 90%, clearly demonstrates the potential of adaptive inference to overcome the limitations of fixed decoding strategies. This significant performance gain suggests that the proposed method can effectively navigate the complex solution space of Sudoku puzzles by prioritizing the most relevant tokens at each step. In summary, the paper's strengths lie in its novel approach to adaptive inference, its grounding in energy minimization theory, and the promising empirical results on Sudoku puzzles. These aspects highlight the potential of the proposed method to advance the field of discrete generative modeling.

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 Weaknesses

After a thorough review of the paper, I have identified several significant weaknesses that undermine its overall impact. Firstly, the paper suffers from a lack of clarity and detail in its methodological description. The 'METHOD' section provides a high-level overview of the adaptive inference strategy but omits crucial implementation specifics. For instance, the paper does not specify the exact architecture of the Masked Diffusion Model used, including the number of layers, hidden dimensions, and other relevant hyperparameters. Similarly, the training procedure is not described in sufficient detail, such as the optimizer used, learning rate schedule, and batch size. This lack of detail makes it difficult to understand the experimental setup and reproduce the results. The description of the adaptive inference strategy itself is also vague. While the paper mentions the use of contextual information and energy minimization, it does not provide a clear explanation of how these components are implemented. The specific algorithms and techniques used to dynamically adjust token ordering are not clearly articulated. This lack of clarity

extends to the 'EXPERIMENTAL SETUP' section, where the evaluation metrics are not clearly defined. While the paper mentions 'solving accuracy and completion time', it does not specify how accuracy is measured (e.g., percentage of completely and correctly solved Sudoku grids). This lack of precision makes it difficult to interpret the results and compare them with other methods. My confidence in this assessment is high, as these omissions are clearly evident in the provided text. Secondly, the paper lacks a formal problem formulation. While the 'motivation' section outlines the general problem of computational inefficiencies and fixed decoding in MDMs, it does not formally define the problem being addressed with specific input/output spaces and assumptions. This absence of a clear problem definition makes it challenging to understand the context and significance of the proposed method. The paper implicitly defines the input and output spaces through the Sudoku experiment, but a more general problem formulation is missing. This lack of clarity makes it difficult to assess the scope and limitations of the proposed approach. My confidence in this assessment is high, as there is no explicit section or clear statement defining the problem being solved. Thirdly, the paper's writing quality is poor, with numerous grammatical errors and unclear sentences. For example, in Line 047, the sentence "Masked difusion models (MDMs) have gained traction for their ability to model complex data distributions in discrete spaces, yet they face significant challnges during training and inference" contains a spelling error ("challnges"). In Line 048, the phrase "computational complexity of their training processes and the limitations of fixed decoding strategies" could be phrased more clearly. In Line 050, the sentence "Existing literature emphasizes the efficacy of autoregressive models (ARMs) for structured tasks,yet fails to explore adaptive inference strategies that exploit the flexibility of MDMs during inference" is somewhat awkward. These examples, among others, demonstrate a lack of attention to detail in the writing, which detracts from the paper's overall clarity and professionalism. My confidence in this assessment is high, as these errors are directly observable in the text. Fourthly, the paper lacks a strong theoretical justification for the proposed method. While the 'method overview' mentions that the strategy is grounded in the theoretical framework of energy minimization, the paper does not provide a formal theoretical analysis, such as convergence guarantees or error bounds. The explanation of the energy minimization framework is conceptual but lacks mathematical rigor. The paper does not provide any theoretical analysis of the proposed method, such as convergence guarantees or error bounds. This lack of theoretical analysis makes it difficult to assess the robustness and reliability of the proposed method. My confidence in this assessment is medium, as the paper does mention the theoretical basis but does not provide any formal proofs. Fifthly, the experimental evaluation is limited to a single dataset, Sudoku puzzles. While Sudoku is a good example of a structured task, evaluating the method on other types of discrete data and tasks would provide stronger evidence of its generalizability. The paper does not evaluate the proposed method on other datasets, such as those used in natural language processing or computer vision. This limited evaluation raises concerns about the applicability of the proposed method to other domains. My confidence in this assessment is high, as the paper explicitly states that Sudoku puzzles were used as the benchmark task. Finally, the paper lacks an ablation study. While the 'experiments' section compares the 'Adaptive MDM' with a 'Standard MDM' and an 'ARM,' this comparison shows the overall impact of the adaptive inference strategy but does not isolate the contribution of individual components within that strategy. There is no systematic evaluation of the impact of the contextual information gathering, the energy minimization framework, or other components of the adaptive inference strategy. This lack of an ablation study makes it difficult to understand which aspects of the proposed method are most critical for its performance. My confidence in this assessment is high, as there is no mention of an ablation study in the paper.

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 Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should significantly expand the 'METHOD' section to include a detailed description of the Masked Diffusion Model architecture, the training procedure, and the adaptive inference strategy. This should include details such as the number of layers, hidden dimensions, activation functions, optimizer used, learning rate schedule, batch size, and the exact algorithm used to dynamically adjust token ordering. The authors should also provide a clear and precise definition of the evaluation metrics used in the experiments. This level of detail is crucial for reproducibility and for a deep understanding of the proposed method. Secondly, the authors should include a formal problem formulation section that clearly defines the input and output spaces, the assumptions made, and the specific problem being addressed. This will provide a clearer context for the proposed method and its significance. Thirdly, the authors should thoroughly revise the paper to improve its writing quality. This should include correcting all grammatical errors, clarifying unclear sentences, and ensuring a consistent and professional tone throughout the paper. Fourthly, the authors should provide a more rigorous theoretical justification for the proposed method. This should include a formal mathematical analysis of the energy minimization framework, along with convergence guarantees and error bounds. This will strengthen the theoretical foundation of the paper and increase confidence in the proposed method. Fifthly, the authors should significantly expand the experimental evaluation to include a wider range of datasets and tasks. This should include datasets from other domains, such as natural language processing and computer vision, to demonstrate the generalizability of the proposed method. The authors should also compare their method with other state-of-the-art methods for solving these tasks. Finally, the authors should conduct a thorough ablation study to evaluate the impact of each component of the adaptive inference strategy. This should include systematically removing or modifying individual components to assess their contribution to the overall performance. This will provide a deeper understanding of the proposed method and its limitations. In addition to these specific recommendations, the authors should also consider adding a more thorough review of related work, positioning their method within the broader context of MDMs and adaptive inference techniques. This will help to highlight the novelty and significance of their contribution. By addressing these weaknesses, the authors can significantly improve the quality and impact of their paper.

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 Questions

After reviewing the paper, I have several questions that I believe are critical for a deeper understanding of the proposed method. Firstly, regarding the adaptive inference strategy, how exactly is the contextual information gathered and used to dynamically adjust the token ordering? The paper mentions using contextual information and energy minimization, but it does not provide specific details on the algorithms and techniques used. What is the computational overhead of this adaptive process, and how does it compare to the computational cost of standard MDM inference? While the paper mentions completion time as a metric, it does not provide a detailed analysis of the computational cost of the adaptive inference strategy. Secondly, regarding the energy minimization framework, how are the local and global potentials defined and calculated? The paper mentions that the energy function is a sum of local and global potentials, but it does not provide specific details on their formulation. What are the specific mathematical equations used to calculate these potentials? How sensitive is the performance of the method to the choice of these potentials? Thirdly, regarding the experimental evaluation, why was Sudoku chosen as the sole benchmark task? While Sudoku is a good example of a structured task, it is not representative of all discrete generative tasks. What are the limitations of evaluating the method only on Sudoku puzzles? What other datasets and tasks would be suitable for evaluating the generalizability of the proposed method? Fourthly, regarding the model architecture, what is the specific architecture of the Masked Diffusion Model used in the experiments? The paper does not provide any details on the model architecture, making it difficult to understand the experimental setup. What are the specific hyperparameters used for training the model? What is the sensitivity of the method to the choice of hyperparameters? Finally, regarding the theoretical justification, while the paper mentions energy minimization as the theoretical basis, it lacks a formal mathematical analysis. What are the convergence guarantees of the proposed method? What are the error bounds? How does the proposed method relate to other existing methods for discrete generative modeling? These questions are crucial for a deeper understanding of the proposed method and its limitations. Addressing these questions will significantly strengthen the paper and increase confidence in its findings.

 **Scores**

Soundness: 1.75

Presentation: 1.25

Contribution: 1.5

Rating: 3.25

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AI Review from ZGCA

AI Review from ZGCA will be automatically processed

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 **Summary**

The paper proposes an adaptive inference strategy for Masked Diffusion Models (MDMs) that dynamically adjusts token ordering during decoding based on contextual information. The goal is to prioritize easier or more informative tokens first to sidestep hard subproblems, drawing a connection to an energy minimization view (Section 3). The main empirical claim is a dramatic improvement on Sudoku: boosting accuracy from less than 7% to approximately 90% (Abstract, Figure 2, Section 5), allegedly outperforming larger autoregressive models. The method is framed as a refinement over prior adaptive unmasking by focusing on which tokens to predict next rather than how many to unmask.

 **Strengths**

- Clear and timely problem motivation: decoding order matters in MDMs for discrete reasoning tasks (Section 1), and adaptive inference is a plausible and impactful lever.
- Conceptual refinement relative to prior adaptive unmasking: emphasis on prioritizing which tokens to predict based on context (Sections 2–3).

- Potentially large practical impact if validated: reported Sudoku accuracy jump to ~90% (Abstract, Figure 2).
- High-level connection to energy minimization (Section 3) is conceptually appealing and aligns with recent analyses of MDMs.

✖ Weaknesses

- Insufficient methodological detail: Section 3 does not specify the scoring function used to prioritize tokens, the search/greedy policy, tie-breaking, lookahead/beam width, stopping criteria, or computational complexity and runtime overhead.
- Baseline is inadequately defined: Section 4 only states a comparison to a 'standard MDM and an ARM under traditional decoding methods' without architecture details, parameter counts, training budgets, datasets/puzzle distributions, or decoding hyperparameters. The flagship '<7%' baseline' is not substantiated.
- Lack of reproducibility information: no hyperparameters, seeds, data splits, or implementation details for either the proposed method or baselines (Section 4).
- Results lack statistical support and analysis depth: ~90% accuracy is reported without variance, confidence intervals, or significance tests; no runtime or compute measurements to support claims of 'sidestepping computationally heavy subproblems' (Abstract, Section 5).
- Figures are not informative: Figure 1 panels are generic loss curves without clear experimental mapping; Figure 2 repeats the headline claim but does not provide evaluation protocol details.
- Clarity issues: multiple typos ('Siku', 'Sodomu') and limited organization make it hard to follow the setup and assess validity.
- Limited scope: evaluation focuses on Sudoku; there is no evidence of generalization to other logic puzzles or broader discrete generative tasks, despite claims in Sections 1 and 6.
- Positioning relative to closely related recent work on token ordering in MDMs is not fully disentangled; originality appears as a refinement rather than a fundamentally new paradigm.

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? Questions

- Algorithmic specifics: How exactly is the next token (or set of tokens) chosen? Please specify the scoring function(s), search policy (greedy vs. beam/lookahead), any calibration or uncertainty estimates used, and tie-breaking rules.
- Energy minimization link: What is the explicit energy functional and how does the adaptive ordering approximate or optimize it at inference? Is there a theoretical guarantee (e.g., monotonic decrease) or empirical proxy that correlates ordering decisions with energy reductions?
- Complexity and runtime: What is the computational overhead of adaptive ordering relative to standard MDM decoding? Please report walltime, number of model calls, and memory for both your method and baselines.
- Baselines: What are the precise architectures, parameter counts, training datasets, and decoding procedures for the 'standard MDM' and ARM baselines? How was the '<7%' baseline obtained? Are there stronger ARM baselines (e.g., with search/constraint propagation) at comparable compute?
- Evaluation protocol: What Sudoku distributions were used (e.g., difficulty levels, generator), how many puzzles, and how were train/validation/test splits created? Are results averaged over multiple seeds with confidence intervals?
- Generalization: Does the method transfer to other logic puzzles (e.g., Kakuro, Nonograms) or to non-puzzle discrete tasks (e.g., code, structured prediction)? Please provide results beyond Sudoku to support broader claims.
- Ablations: Can you ablate the ordering heuristic (entropy, confidence, constraint-violation estimate, marginal improvement proxy, etc.), the number of tokens selected per step, and any lookahead depth? How sensitive is performance to these choices?
- Comparison to adaptive unmasking: How does your 'which-token' prioritization compare to methods that adapt 'how many' tokens to unmask per step when both are properly tuned under identical budgets?
- Constraint handling: Do you use explicit Sudoku constraint checks or post-hoc repair during decoding? If so, please detail the interface; if not, how do you ensure correctness without external solvers?
- Reproducibility: Will you release code, model checkpoints, and the exact puzzle datasets to support verification?

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⚠ Limitations

- Method is currently under-specified; reproducibility is not possible from the present text.
- Evidence is limited to Sudoku; generalization to other tasks is unproven.
- Potential computational overhead at inference (due to scoring/selection) is not measured; scaling to longer sequences or larger boards may be challenging.
- Reliance on task-specific heuristics or constraint checks (if used) could reduce generality; not discussed.
- No error analysis to understand failure modes or robustness across puzzle difficulties.
- Potential negative societal impact appears minimal for logic puzzles, but if extended to safety-critical domains, overconfident decoding without uncertainty calibration could be risky.

🖼 Image Evaluation

Cross-Modal Consistency: 32/50

Textual Logical Soundness: 16/30

Visual Aesthetics & Clarity: 12/20

Overall Score: 60/100

Detailed Evaluation (≤500 words):

Visual ground truth (Image-First):

- Figure 1(a): Baseline Loss Curves — line plots of training/validation loss vs epochs; both decrease smoothly.
- Figure 1(b): Activation Ablation — two separate plots: training (blue) and validation (green); single series each with visible phase jumps ~50 and ~100 epochs; no legend.
- Figure 1(c): Dataset Variation — three plots titled Easy/Medium/Hard; training and validation losses both decline; gaps shrink with epochs.
- Figure 1(d): Regularization — two plots “Without L2” and “With L2”; both show decreasing losses; with-L2 validation slightly lower.
- Figure 2: Sudoku accuracy improvement — text-only graphic: “Baseline <7%” vs “Adaptive ≈90%”; no axes or sample counts.

1. Cross-Modal Consistency

- Major 1: Fig. 2 does not visualize data (no axes/sample size), yet Sec. 5 relies on it to evidence the core gain. Evidence: “Our experiments revealed... as illustrated in Figure 2... ≈ 90%.”
- Minor 1: Sub-figure labeling inconsistent; several panes lack (b1/b2, etc.) tags, risking misreference. Evidence: Fig. 1(b–d) images titled without pane tags.
- Minor 2: Task name inconsistencies (“Siku”, “Sodomu” vs Sudoku). Evidence: Abstract “Siku”; Sec. 5 “Sodomu”.
- Minor 3: “Activation ablation” implies comparison but shows single-series curves. Evidence: Fig. 1(b) shows one curve per plot.

2. Text Logic

- Major 1: Unsupported flagship claim of beating ARMs; no quantitative comparison table/figure. Evidence: Abstract “even outperforming autoregressive models...”.
- Major 2: Method lacks actionable algorithm for adaptive ordering (selection rule, scoring, complexity), blocking reproducibility. Evidence: Sec. 3 “dynamically adjust token ordering... prioritize token predictions”.
- Minor 1: Evaluation protocol under-specified (dataset size/difficulty mix, timing metric, seeds). Evidence: Sec. 4 “focused on Sudoku... measuring... accuracy and completion time”.
- Minor 2: Several typos reduce clarity. Evidence: Sec. 5 “Sodomu solving accuracy”.

3. Figure Quality

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- Major 1: Fig.1 caption too generic; readers can't infer which settings each pane represents. Evidence: "Figure 1: Performance evaluation across various experiments."
- Minor 1: Fig. 1(b) lacks legend specifying which activations or switch points; fails Figure-Alone test. Evidence: Titles only ("Activation Ablation Training/Validation Loss").
- Minor 2: Fig. 2 is text-only; add bar plot with CIs and N. Evidence: Fig. 2 content.

Key strengths:

- Clear learning curves in (a), (c), (d) suggest stable optimization.
- Focus on adaptive inference for discrete MDMs is timely and potentially impactful.

Key weaknesses:

- Central ARM comparison absent.
- Method description too high-level to reproduce.
- Fig. 2 lacks substantive evidence; sub-figure labeling and captions need tightening.
- Typos and inconsistent terminology (Sudoku) hurt polish.

 Scores

Originality: 2

Quality: 1

Clarity: 1

Significance: 2

Soundness: 1

Presentation: 2

Contribution: 2

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Rating: 3

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AI Review from SafeReviewer

 AI Review from SafeReviewer will be available soon

 Summary

This paper introduces an adaptive inference strategy for Masked Diffusion Models (MDMs), aiming to enhance their performance on discrete generative tasks, specifically Sudoku solving. The core idea revolves around dynamically adjusting the token ordering during the decoding phase, rather than relying on a fixed sequence. The authors propose that this adaptive approach allows the model to prioritize easier tokens first, effectively sidestepping computationally intensive subproblems and leading to improved accuracy. The method is grounded in the theoretical framework of energy minimization, suggesting that optimal token arrangements can minimize generation errors. Empirically, the authors demonstrate that their adaptive inference strategy significantly boosts Sudoku solving accuracy, increasing it from less than 7% to approximately 90%. They also claim that this performance surpasses that of autoregressive models (ARMs) with significantly more parameters. The paper positions its contribution as addressing the underutilization of MDMs' flexibility during inference, contrasting it with the established efficacy of ARMs in structured tasks. While the paper highlights the potential of adaptive inference, it also acknowledges the computational challenges associated with training MDMs. The overall significance of the work lies in its exploration of adaptive inference as a means to enhance the practical applicability of MDMs in complex generative tasks, particularly those involving structured data. However, the paper's brevity and lack of detailed explanations in several key areas limit the ability to fully assess the novelty and robustness of the proposed method.

✔ Strengths

The core strength of this paper lies in its introduction of an adaptive inference strategy for Masked Diffusion Models (MDMs), which demonstrates a significant improvement in performance on the Sudoku solving task. The empirical results, as stated in the abstract and experiments section, show an increase in solving accuracy from less than 7% to approximately 90%. This substantial improvement suggests that the proposed method effectively leverages the flexibility of MDMs during inference. Furthermore, the paper claims that this performance surpasses that of autoregressive models (ARMs) with significantly more parameters, highlighting the potential efficiency gains of the adaptive approach. The paper's focus on addressing the underutilization of MDMs' flexibility during inference is a valuable contribution, as it shifts the focus from the computational challenges of training to the potential of optimizing the inference process. The grounding of the method in the theoretical framework of energy minimization, as mentioned in the method section, provides a theoretical basis for the adaptive approach, suggesting that optimal token arrangements can minimize generation errors. While the paper's brevity limits the depth of explanation, the core idea of dynamically adjusting token ordering based on contextual information is intuitively appealing and aligns with the need for more efficient inference strategies in generative models. The paper's focus on a practical application, Sudoku solving, provides a clear and tangible demonstration of the method's potential. Despite the limitations in detail, the empirical achievement of a substantial accuracy increase is a notable strength of this work.

✖ Weaknesses

After a thorough review, I've identified several significant weaknesses in this paper, primarily stemming from its lack of detail and clarity. Firstly, the paper suffers from a severe lack of methodological detail. The 'Method' section is extremely concise, providing only a high-level overview of the adaptive inference strategy. Specifically, the paper states that the method "leverages contextual information to prioritize token predictions," but it fails to explain *how* this prioritization is achieved. There is no description of the algorithm used to determine the optimal token ordering, nor are there any mathematical formulations or pseudo-code provided. This lack of detail makes it impossible to understand the core mechanics of the proposed method and, crucially, to reproduce the results. The absence of reproducibility is a cornerstone of scientific research. My confidence in this assessment is very high, as the absence of these fundamental details is evident throughout the 'Method' section. Secondly, the experimental evaluation is severely limited. The 'Experiments' section focuses solely on Sudoku puzzles across various datasets or tasks presented. While Sudoku is a useful benchmark for structured reasoning, it is insufficient to demonstrate the generalizability of the proposed adaptive inference strategy. The paper lacks experiments on other types of logic puzzles or any other discrete generative tasks. This narrow focus significantly weakens the paper's claims about the broader applicability of the method. My confidence in this assessment is high, as the 'Experiments' section explicitly states that "To evaluate our adaptive inference strategies, we focused on Sudoku puzzles as a benchmark task," and no other tasks are mentioned. Thirdly, the paper lacks crucial details about the model architecture and training process. The paper mentions using a "standard MDM" as a baseline but provides no information about its specific architecture, such as the number of layers, hidden dimensions, or attention mechanisms. Similarly, the training procedure is not described, leaving the reader with no understanding of how the MDM was trained. This lack of information makes it difficult to assess the validity of the results and to compare them to other approaches. My confidence in this assessment is high, as the 'Method' and 'Experiments' sections lack any details about the model or training process. Fourthly, the paper does not provide sufficient information about the computational resources used for the experiments. The paper mentions "completion time" as a metric but does not provide any specific timing results or details about the hardware used for the experiments. This lack of information makes it difficult to assess the practical efficiency of the proposed method. My confidence in this assessment is high, as the 'Experiments' section mentions "completion time" but provides no concrete data or hardware specifications. Fifthly, the paper's writing style is overly brief and lacks the necessary detail expected in an academic paper. The descriptions are often high-level and lack the necessary detail to fully understand the proposed method and its evaluation. This brevity makes it difficult to follow the authors' reasoning and to assess the validity of their claims. My confidence in this assessment is high, as the short length of the 'Method' and 'Experiments' sections and the lack of detailed explanations are readily apparent. Finally, the paper does not adequately discuss the limitations of the proposed method. While the 'Conclusion' section mentions future work, it does not explicitly address the limitations of the current approach. This omission is a significant weakness, as it prevents a balanced assessment of the method's strengths and weaknesses. My confidence in this assessment is high, as there is no dedicated section or substantial discussion on the limitations of the proposed method. In summary, the paper's lack of methodological detail, limited experimental evaluation, missing information about the model and training process, absence of computational resource details, brevity, and lack of discussion on limitations are all significant weaknesses that undermine the paper's credibility and impact.

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💡 Suggestions

To address the identified weaknesses, I recommend several concrete and actionable improvements. Firstly, the authors must significantly expand the 'Method' section to provide a detailed explanation of the adaptive inference strategy. This should include a clear description of the algorithm used to determine the optimal token ordering, along with the mathematical formulations and pseudo-code. The authors should also explain how contextual information is used to prioritize token predictions. This level of detail is crucial for reproducibility and for understanding the core mechanics of the proposed method. Secondly, the authors should broaden the experimental evaluation to include a wider range of datasets and tasks. In addition to Sudoku puzzles, they should consider including other types of logic puzzles, as well as other discrete generative tasks. This would provide a more comprehensive assessment of the generalizability of the proposed adaptive inference strategy. Thirdly, the authors must provide detailed information about the model architecture and training process. This should include specifics about the number of layers, hidden dimensions, attention mechanisms, and any other relevant architectural details. The training procedure should also be described in detail, including the optimization algorithm, learning rate, batch size, and any other relevant hyperparameters. This information is essential for understanding the experimental setup and for comparing the proposed method with other approaches. Fourthly, the authors should include specific details about the computational resources used for the experiments. This should include information about the hardware used, such as the type of CPU and GPU, as well as the amount of memory. They should also provide specific timing results for the experiments, allowing for a more thorough assessment of the method's efficiency. Fifthly, the authors should revise the writing style to be more detailed and informative. The descriptions should be more thorough, and the reasoning should be clearly explained. The authors should avoid overly brief statements and provide sufficient context for the reader to understand the proposed method and its evaluation. Finally, the authors should include a dedicated section or substantial discussion on the limitations of the proposed method. This should include a balanced assessment of the method's strengths and weaknesses, as well as a discussion of potential areas for future research. By addressing these points, the authors can significantly improve the quality and impact of their work. These changes are within the scope of a revised paper and are essential for addressing the identified weaknesses.

? Questions

After reviewing the paper, I have several questions that I believe are crucial for clarifying the methodology and understanding the results. Firstly, I am curious about the specific algorithm used to determine the optimal token ordering. The paper mentions leveraging contextual information, but it does not provide any specific details about how this information is used to prioritize token predictions. What is the precise mechanism for selecting the next token to predict? Is there a scoring function involved? Secondly, I am interested in the computational cost of the adaptive inference strategy. While the paper claims that the method is efficient, it does not provide any specific timing results or details about the computational resources used. How does the computational cost of the adaptive inference strategy compare to a standard MDM with a fixed decoding order? Is there a trade-off between accuracy and computational cost? Thirdly, I am curious about the generalizability of the proposed method to other discrete generative tasks. The paper focuses solely on Sudoku puzzles, but it is unclear whether the adaptive inference strategy would be effective on other types of logic puzzles or other discrete data. What are the limitations of the proposed method in terms of its applicability to different tasks? Fourthly, I would like to know more about the training process of the MDM used in the experiments. The paper mentions using a "standard MDM" but provides no details about its architecture or training procedure. What are the specific architectural details of the MDM? What optimization algorithm and hyperparameters were used during training? Finally, I am interested in the sensitivity of the proposed method to different hyperparameters. Does the performance of the adaptive inference strategy depend on the choice of hyperparameters? Are there any specific hyperparameters that are particularly important for achieving good results? These questions are aimed at clarifying the core methodological choices and assumptions of the paper, and I believe that answering them would significantly enhance the transparency and credibility of the work.

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📊 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Confidence: 4.75

Rating: 1.5

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights

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